

Mini E-Commerce Backend

Orders Module Blueprint

1. Orders Module Overview

Responsibilities:

- Checkout
- Create Order
- Get User Orders
- Get Order By ID
- Update Order Status (Admin)
- Calculate Total Price

2. Orders APIs

Checkout

Item	Value
Method	POST
Endpoint	/orders
Authorization	User
Response	201 Created

Response

```
{  
  "message": "Order created successfully"  
}
```

Get User Orders

Item	Value
Method	GET
Endpoint	/orders
Authorization	User
Response	200 OK

Get Order By ID

Item	Value
Method	GET
Endpoint	/orders/:id
Authorization	User
Response	200 OK

Update Order Status

Item	Value
Method	PATCH
Endpoint	/orders/:id
Authorization	Admin
Response	200 OK

Request

```
{  
  "status": "Delivered"  
}
```

3. Checkout Flow

Client → JWT Middleware → Get User Cart → Calculate Total Price → Create Order → Save Order → Clear Cart → Response

4. Order Lifecycle

Cart → Checkout → Order Created → Processing → Shipped → Delivered

5. Order Operations

Checkout

- Read Cart
- Calculate Total
- Create Order
- Save Order
- Clear Cart

Get Orders

Return all user orders.

Get Order By ID

Return a single order.

Update Status

Admin updates order status.

6. Business Rules

- User can only access his own orders.
- Admin can update order status.
- Order cannot be created if the cart is empty.
- Total Price is calculated before saving.
- Cart is cleared after successful checkout.

7. Folder Structure

```
controllers/  
  order.controller.js
```

```
routes/  
  order.route.js
```

8. Controller Responsibilities

Function	Responsibility
checkout	Create Order
getOrders	Return User Orders
getOrder	Return Single Order
updateStatus	Update Order Status

9. HTTP Status Codes

Code	Description
200	Success
201	Order Created
400	Empty Cart
401	Unauthorized
403	Forbidden
404	Order Not Found
500	Internal Server Error

10. Common Error Scenarios

Error	Status Code
Cart is Empty	400
Order Not Found	404
Unauthorized User	401
Forbidden Access	403

11. Testing Checklist

- Checkout
- Get Orders
- Get Order By ID
- Update Order Status
- Empty Cart
- Calculate Total Price
- JWT Authentication
- Authorization

12. Interview Notes

Why calculate total before saving?

To store the final amount paid at purchase.

Why clear cart?

Products were converted into an order.

Why only own orders?

To protect user privacy.

Why only Admin updates status?

Order processing is managed by administrators.

What if cart is empty?

400 Bad Request

Why save product price?

To preserve the original purchase price if it changes later.

13. Sequence Diagram

Client → POST /orders → JWT Middleware → Order Controller → Read Cart
→ Calculate Total Price → Create Order → Save Order → Clear Cart →
Return Response